Clustering Part I

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J&M: 9.4.1

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Why Cluster

- Group Similarity
- Compression algorithms
- Soundscape analysis

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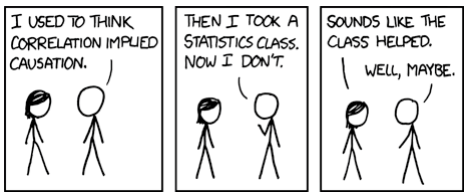
Clustering Methods

- K-means/Vector Quantization
- Gaussian Mixed Models

CAUTION!

Ecologist/Marine Biologist/ Acoustician/ Statistician crossing



xkcd.com

Some Background: Similarity

- How similar are two vectors?
- Distance metric (distortion)
  - $d(x,y) = \begin{cases} 0 & x = y \\ > 0 & x \neq y \end{cases}$
  - Satisfies triangle $\neq$: $d(x,y) + d(y,z) \geq d(x,z)$
  - Symmetric: $d(x,y) = d(y,x)$

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Distance Between Points

- One Dimension (x_1, y_1)
- Two Dimensions (x_1, y_1, x_2, y_2)
- N Dimensions?

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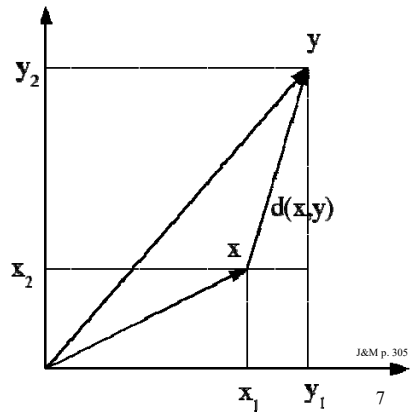
Euclidean distance/distortion

Straight line distance (squared) between two points

$$d^2(x,y) = \sum_{i=1}^{D} (x_i - y_i)^2$$

as a vector operation:

$$d^2(x,y) = (\text{x-y})^T (x - y)$$



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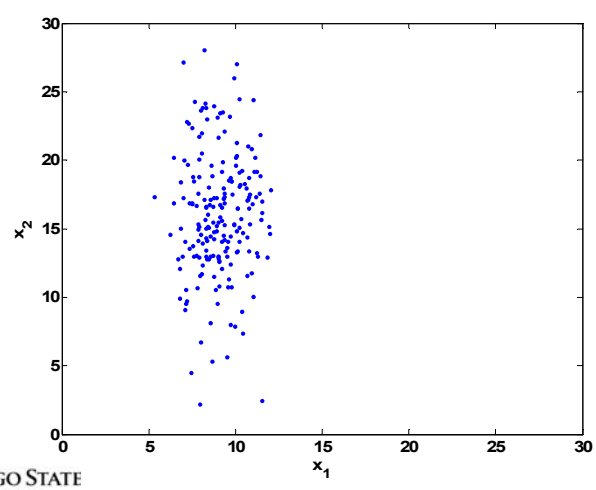
Does Euclidean distance always make sense?



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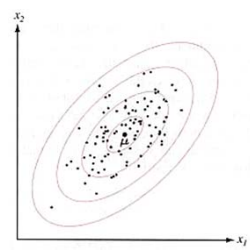
Distortion - Mahalanobis

- Mahalanobis distortion
  - Accounts for the variance and covariance (Σ)
  - Removes assumption of equal scaling

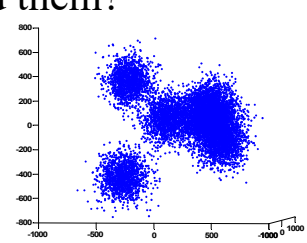


$$d_{Mahalanobis}(\vec{x},\vec{y}) = (\vec{x}-\vec{y})^t \Sigma^{-1} (\vec{x}-\vec{y})$$

k-means clustering

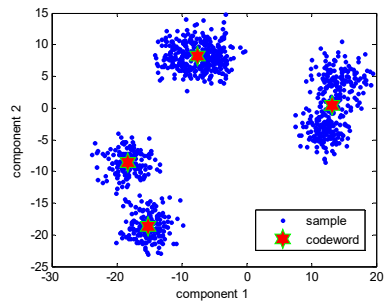
also known as vector quantization

- Let's assume that we know there are *k* clusters.
- How do we find them?

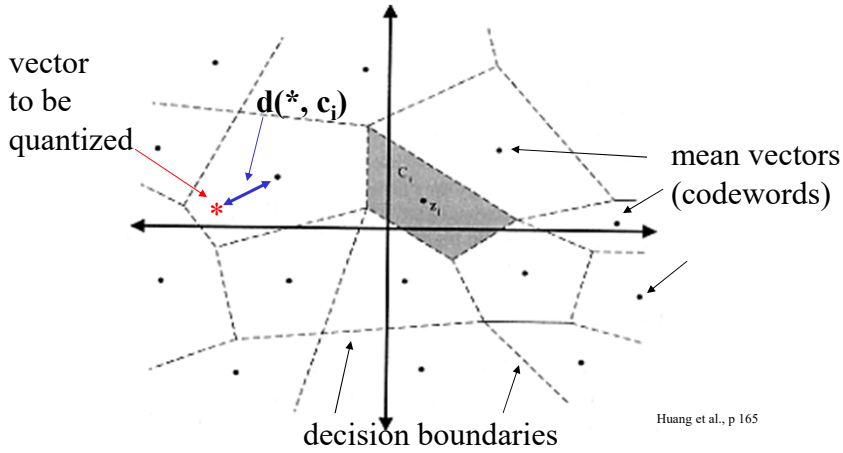


k-means clustering

- Find vectors representative of clusters.
- Representative vectors sometimes called codewords and the collection a codebook.



R^2 partition induced by *k*-means



Huang et al., p 165

k-means/Vector Quantization clustering

Select *k* vectors at random as initial centers from training sample X
done = false;
old_distortion = ∞
while not done
    Compute $d(x_i, c_j)$ for each training vector and center
    Partition training vectors according to c_j which produced smallest distortion
    Compute new centers by taking the mean (centroid) of each partition
    distortion = compute avg. minimum distortion for all training vectors
    done = distortion / old_distortion > threshold
    old_distortion = distortion

Quantizing

Quantization finds the closest codeword in codebook *c*:

$$q(\vec{x}, c) = \vec{c}_i \leftrightarrow i = \arg\min_{1 \le k \le K} d(\vec{x}, \vec{c}_k)$$

Sometimes we want the distortion to the closest codeword:

$$\text{distortion}(\vec{x}, c) = \min_{1 \le k \le K} d(\vec{x}, \vec{c}_k)$$

Using Vector Quantization/k-means

- Unsupervised classifier
  - Centroids represent the distribution of items
  - Used in discrete hidden Markov models
- Supervised classifiers
  - Construct codebooks for each class
  - Find class with minimum distortion

A Supervised VQ classifier

- Training

  For each class ω_i in Ω construct a codebook.
- Testing

  Given a set of test vectors $X = \{\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_T\}$

  Find codebook with smallest distortion for the vectors.

VQ Classification

$MinDistortion = \infty$

for $cidx$ = 1 to $|\Omega|$

  $SumDistortion = 0$

  for $vidx$ = 1 to T

    $SumDistortion = SumDistortion + distortion(\vec{x}_{vidx}, book_{cidx})$

  if $SumDistortion < MinDistortion$

    $MinDistortion = SumDistortion$

    $MinIdx = cidx$

Decide that X belongs to class ω_{MinIdx}

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Note: Frequently, the average distortion is computed.

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Clustering Part II

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J&M: 9.4.2 to end of 9.4

supplemental readings from

Huang, X., A. Acero, H.W. Hon (**2001**). ***Spoken Language Processing. Prentice Hall PTR, Upper Saddle River, NJ***

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What happen when distributions overlap?

- We need a different way to cluster our data
- Soft methodology- probabilities not certainties

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Multidimensional Gaussians

$$f_i(\bar{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma_i|^{1/2}} e^{\left(\frac{-1}{2}(\bar{x}-\mu_i)^t \Sigma_i^{-1}(\bar{x}-\mu_i)\right)}$$

where d =# dimensions, i = classifier index

Maximum likelihood estimators are simply sample mean & variance

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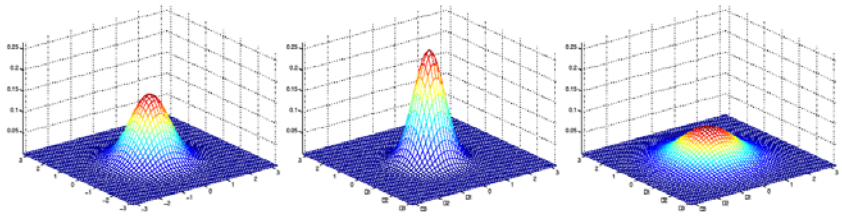
Gaussian Intuitions: Size of Σ



- $\mu = [0\ 0]$ $\mu = [0\ 0]$ $\mu = [0\ 0]$
- $\Sigma = I$ $\Sigma = 0.6I$ $\Sigma = 2I$
- As Σ becomes larger, Gaussian becomes more spread out; as Σ becomes smaller, Gaussian more compressed

Text and figures from Andrew Ng's lecture notes for CS229 (via Jurafsky & Martin)

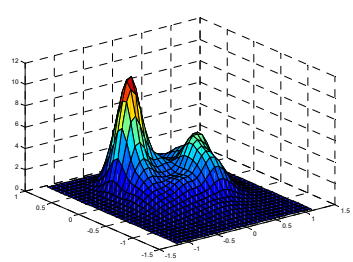
Gaussian Mixture Models (GMMs)

- Another unsupervised learner
- Similar to *k*-means, multiple GMMs can be used for supervised classification
- N normal distributions represent data's distribution.
  - Means: Similar to the codewords
  - $P(x \mid \mu, \sum)$ measures similarity

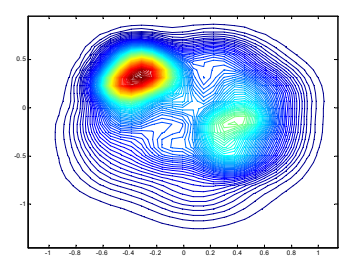
GMM: Parameters

- Parameters for each of K mixtures k=1...K:
  - μ_k mean
  - $\sum_k$ variance-covariance matrix
  - c_k mixture weight where $\sum_{k=1}^{K} c_k = 1$
  - Use Φ_k to denote (c_k, μ_k, $\sum_k$) and Φ to denote the entire set of parameters (Note: Authors do not include c_k in Φ_k)

Sample 16 mixture model Based upon R^2 cepstral speech data

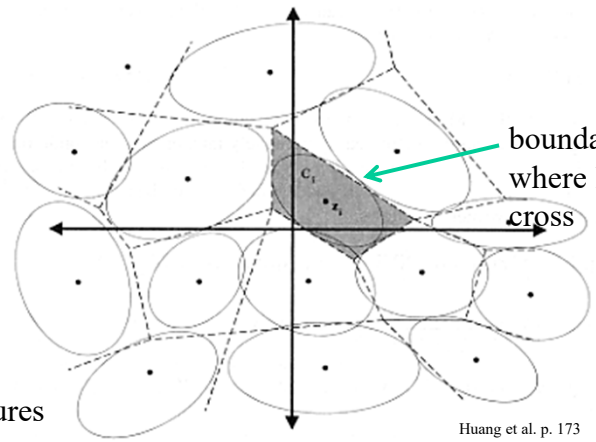


Surface plot of pdfs



Equal likelihood contour lines

Unsupervised partitions



Huang et al. p. 173

different distribution than the previous slide

GMM: Evaluation probability

- Probability of an observation:

$$\Pr(\vec{x} \mid \Phi) = \sum_{k=1}^{K} c_k N_k(\vec{x} \mid \mu_k, \Sigma_k)$$

$$= \sum_{k=1}^{K} c_k \frac{1}{(2\pi)^{d/2} \left|\Sigma_k\right|^{1/2}} e^{\left(\frac{-1}{2}(\vec{x}-\mu_k)^t \Sigma_k^{-1} (\vec{x}-\mu_k)\right)}$$

GMM: Mixture weights

- Interpretation of mixture weights c_i
  - Prior probability that observation x comes from mixture i.

  ... or ...

  - Scaling of mixtures such that all mixtures together form a pdf.

GMM Parameter estimation: The EM algorithm

- Suppose that we wish to maximize a parameter set φ given Y=y, but φ also depends upon random variable X.
- That is, if we had X=x, we could select φ to maximize:

$$P(X = x, Y = y \mid \phi)$$

The EM Algorithm

- X=x is unavailable, and we will refer to it as hidden.
- Outline of the EM algorithm:
  1. Use an initial estimate of φ to determine E[X] taking into account Y=y.
  2. Use Y=y and E[X] to determine a new φ.
  3. If converged, stop, otherwise goto 1.
- Convergence is guaranteed

GMM: Estimation

- Application of the EM algorithm
- Expectation step
  - Determine how well each mixture models each observation
    - Determine probability of observation with respect to mixture in question.
    - Divide by probability of seeing the observation (sum across mixtures).

GMM: Expectation step

- How well does mixture k model y_i?

$$\gamma_k^i = \frac{P(y_i \mid \text{mixture } k)}{P(y_i \mid \text{all mixtures})} = \frac{P(y_i \mid \Phi_k)}{P(y_i \mid \Phi)} = \frac{c_k \, P(y_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} c_j \, P(y_i \mid \mu_j, \Sigma_j)}$$

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GMM: Expectation step

- Also need to determine how well each mixture represents the training data
  - Sum the γ's for each mixture over all observations.
  - This is not a probability. Used as a normalizing factor in the maximization step.

$$\gamma_k = \sum_{i=1}^{N} \gamma_k^i$$

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GMM: Maximization step

- $\hat{c}_k = \dfrac{\text{How well mixture k represents training data}}{\text{How well all mixtures represent training data}}$

$$= \frac{\gamma_k}{\sum_{j=1}^{K} \gamma_j}$$

$$= \frac{\gamma_k}{N}$$

GMM: Maximization step

- Why does $\sum_{j=1}^{K} \gamma_j = N$?
  - γ_k represents contribution of kth mixture to probability measured for each observation.
  - The total contribution to the probability for a single observation must be one. (all probability must be accounted for).
  - As there are N observations, the sum is N.

GMM: Maximization step

- $$\hat{\mu}_k = \frac{\sum_{i=1}^{N} \left(\text{Contribution mixture k}\right) \text{ y}_i}{\text{Contribution mixture k across y's}}$$

$$= \frac{\sum_{i=1}^{N} \gamma_k^i y_i}{\sum_{i=1}^{N} \gamma_k^i}$$

$$= \frac{\sum_{i=1}^{N} \frac{c_k \, \mathrm{P}(y_i \mid \mu_k, \Sigma_k)}{\mathrm{P}(y_i \mid \Phi)} y_i}{\sum_{i=1}^{N} \frac{c_k \, \mathrm{P}(y_i \mid \mu_k, \Sigma_k)}{\mathrm{P}(y_i \mid \Phi)}}$$

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GMM: Maximization step

- $$\hat{\Sigma}_k = \frac{\sum_{i=1}^{N} \left(\text{Contribution mixture k}\right) \text{ y}_i\text{'s contribution to cov}}{\text{Contribution mixture k across y's}}$$

$$= \frac{\sum_{i=1}^{N} \gamma_k^i (y_i - \mu_k)(y_i - \mu_k)'}{\sum_{i=1}^{N} \gamma_k^i}$$

$$= \frac{\sum_{i=1}^{N} \frac{c_k \, \mathrm{P}(y_i \mid \mu_k, \Sigma_k)}{\mathrm{P}(y_i \mid \Phi)} (y_i - \mu_k)(y_i - \mu_k)'}{\sum_{i=1}^{N} \frac{c_k \, \mathrm{P}(y_i \mid \mu_k, \Sigma_k)}{\mathrm{P}(y_i \mid \Phi)}}$$

Assumes y_i is a column vector.
Use $(y_i - \mu_k)'(y_i - \mu_k)$ if y_i is a row vector.

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GMM

- Convergence is typically fast 5-15 iterations
- Initialization
  - Train a K codeword VQ codebook
  - For each codeword k, let $X^{(k)}$ denote the training vectors associated with it. Then:

$$c_k = \left|X^{(k)}\right| \Big/ \sum_{j=1}^{K} \left\|X^{(j)}\right\| \qquad \mu_k = mean\left(\left|X^{(k)}\right|\right) = \text{codeword}_k$$

$$\Sigma_k = \text{cov}\left(\left|X^{(k)}\right|\right)$$

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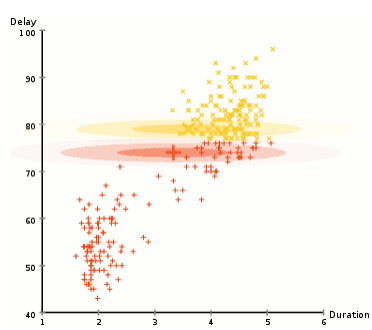
Old Faithful Data



https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm

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Application of GMMs

- Similar to *k*-means/VQ, we can use in unsupervised/supervised manner
- Supervised training of models is similar to our VQ classifier.
  - Joint likelihood instead of average distortion
  - Look for max likelihood instead of min distortion

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GMM Classification

$MaxLikelihood = -\infty$

for $gidx = 1$ to $|\Omega|$

  $JointLikelihood = 0$

  for $vidx = 1$ to T

    % Assume vectors independent

    $JointLikelihood = JointLikelihood + \log\left(P(\vec{x}_{vidx} \mid \phi_{gidx})\right)$

  if $JointLikelihood > MaxLikelihood$

    $MaxLikelihood = JointLikelihood$

    $MaxIdx = gidx$

Decide that X belongs to class ω_{MaxIdx}

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Questions?

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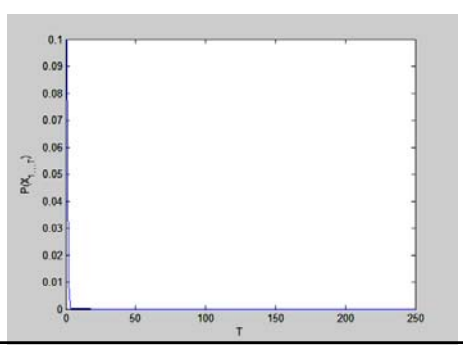
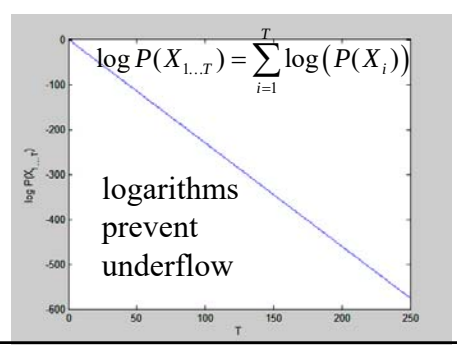
Practicalities

- Consider a set of independent observations

$$P(X_{1\ldots T}) = \Pi_{i=1}^{T} P(X_i)$$

and suppose that the $P(X_i)$ was bounded by .1





Variance-Covariance Matrices

- It is common to assume that Σ_i^{-1} is diagonal.
- Why?

Precomputation

$$f_i(\bar{x}) = \underbrace{\frac{1}{(2\pi)^{d/2}\left|\Sigma_i\right|^{1/2}}}_{\text{constant k: independent of } \bar{x}} e^{\left(\frac{-1}{2}(\bar{x}-\mu_i)^t \Sigma_i^{-1}(\bar{x}-\mu_i)\right)}$$

$$f_i(\bar{x}) = ke^{\left(\frac{-1}{2}(\bar{x}-\mu_i)^t \Sigma_i^{-1}(\bar{x}-\mu_i)\right)} \quad \text{define } \Delta = (\bar{x}-\mu_i)$$

$$\log f_i(\bar{x}) = \log k + \frac{-1}{2}\Delta^t \Sigma_i^{-1}\Delta$$

Σ_i^{-1} can be precomputed, and if it is diagonal $\Sigma_i^{-1}(j,j) = 1/\Sigma_i(j,j)$

and $\Delta^t \Sigma_i^{-1}\Delta$ is just the sum of the products $\sum_{j=1}^{d} \Delta_j \Sigma_i^{-1}(j,j)\Delta_j$